

RAPID-MOC time series April 2004 to March 2024 (v2024.1a)

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Thank you for downloading the RAPID time series of the **Atlantic Meridional Overturning Circulation at 26.5°N**. The data files are updated from time to time, usually when time series can be extended and occasionally when changes are made to the processing of data. If you would like to be informed when changes are made please email the Principal Investigator.

Data policy

Data from RAPID MOC monitoring project are made freely available to the public. Continued funding of this project depends on us being able to justify the usefulness of the data to the Natural Environment Research Council. The project scientists would appreciate it if you would use the data DOI (details below) and add the following acknowledgment to any publications that use this data:

"Data from the RAPID MOC monitoring project are funded by the Natural Environment Research Council, the National Science Foundation (NSF) with support from NOAA. They are freely available from www.rapid.ac.uk/."

Citations

Details of the calculation of the AMOC from the RAPID data are given in the following publications (and the online supporting material).

McCarthy, G.D., and Co-authors, 2015: Measuring the Atlantic Meridional Overturning Circulation at 26°N, *Prog. Oceanogr.*, 130, 91–111. doi:10.1016/j.pocean.2014.10.006.

Cunningham, S. A. and Coauthors, 2007: Temporal variability of the Atlantic Meridional Overturning Circulation at 26.5°N. *Science*, **317**, 935–938.

Kanzow, T. and Coauthors, 2007: Observed flow compensation associated with the MOC at 26.5°N in the Atlantic. *Science*, **317**, 938–941, doi:10.1126/science.1141293.

Digital Object Identifier (DOI) for this release

The full DOI for this data set is: doi: 10.5285/48d0bf43-0598-ceb2-e063-7086abc062f1

The data citation that should be used in journals is: Moat B.I.; Smeed D.A.; Rayner D.; Johns W.E.; Smith, R.; Volkov, D.; Elipot S.; Petit T.; Kajtar J.; Baringer M. O.; and Collins, J. (2026). Atlantic meridional overturning circulation observed by the RAPID-MOCHA-WBTS (RAPID-Meridional Overturning Circulation and Heatflux Array-Western Boundary Time Series) array at 26N from 2004 to 2024 (v2024.1a), British Oceanographic Data Centre - Natural Environment Research Council, UK. doi: 10.5285/48d0bf43-0598-ceb2-e063-7086abc062f1

Changes made when releasing data

January 2026

The timeseries of overturning in depth space is the same as that released in September 2024, but the overturning in density space and other data products are modified. In particular, the calculation of overturning in density space is also made using potential density referenced to 2000dbar, which provides a better representation of the deeper levels of the overturning stream function.

September 2025

The time series is extended to 27th March 2024.

Between the 25th February 2023 and 19th July 2023 the WB2 mooring on the Western boundary was not deployed and was replaced with data from the WB3 mooring.

The Florida Straits transport estimation now uses pressure gauge and altimetry data to fill gaps when cable data are not available. In particular, there has been no cable data since November 2023.

Each time the time series is updated the series mean and standard deviation change. These statistics affect interpolation and quality control of the data and so cause some minor changes to the calculated MOC.

April 2025

The timeseries is the same as that released in September 2024, but additional data products are now included in two additional files. These new files are

- a) Net CDF file at 10-day resolution depth-longitude sections at 26.5°N of:
 - Conservative temperature
 - Absolute salinity
 - Meridional velocity
 - Meridional Ekman velocity
 - Net meridional velocity
- b) NetCDF file of time series with 10-day resolution
 - MOC evaluated in depth space (the same as in other files but on 10-day resolution)
 - MOC evaluated in potential density coordinates
 - Meridional heat transport
 - Freshwater transport
 - Stream function evaluated in depth space
 - Stream function evaluated in potential density coordinates

September 2024

The time series is extended to 11th February 2023.

The deep eastern boundary moorings were removed in February 2022. Since then, monthly climatology created from the existing eastern boundary mooring measurements (between 2004 and 2022) have replaced these data.

We have used the updated Florida Straits transport data which includes a correction for the secular change in the geomagnetic field. The data are available from:

(https://www.aoml.noaa.gov/phod/floridacurrent/data_access.php)

The coefficients used for extrapolation above the shallowest instruments have been updated. An ODO MicroCAT on the eastern boundary, included in the last release, was removed from the T/S field as it was duplicating measurements from an existing MicroCAT at a similar pressure level.

Each time the time series is updated the series mean and standard deviation change. These statistics affect interpolation and quality control of the data and so cause some minor changes to the calculated MOC.

September 2023

The time series is extended to 15th February 2022.

The mid Atlantic Ridge moorings were removed in December 2020. Since then, monthly climatology created from the mid-Atlantic ridge moorings (between 2004 and 2020) has replaced these data.

The coefficients used for extrapolation above the shallowest instruments have been updated.

Each time the time series is updated the series mean and standard deviation change. These statistics affect interpolation and quality control of the data and so cause some minor changes to the calculated MOC.

September 2022

The time series is extended to 15th December 2020.

Each time the time series is updated the series mean and standard deviation change. These statistics affect interpolation and quality control of the data and so cause some minor changes to the calculated MOC.

September 2021

The time series is extended to 9th March 2020.

The top of the WB2 western Boundary mooring failed between 3rd September 2018 and 21st November 2018. During this period the backup WB3 mooring replaced WB2 and the western boundary wedge was extended out to WB3.

Between the 1st September 2019 and 15th March 2020 the deep EBHi mooring on the Eastern boundary failed and was replaced with monthly climatology.

The coefficients used for extrapolation above the shallowest instruments have been updated.

There are minor changes due to a modification in the interpolation used to fill gaps in the Florida Strait cable data.

Each time the time series is updated the series mean and standard deviation change. These statistics affect interpolation and quality control of the data and so cause some minor changes to the calculated MOC.

July 2020

ERA5 wind stress at 26.5°N is now used to calculate the Ekman transport. This has resulted in a decrease of the mean Ekman transport of 0.8 Sv compared with the previous data release. The time series has not been extended past the 3rd September 2018.

June 2019

The time series is extended to 3rd September 2018.

The coefficients used for extrapolation above the shallowest instruments have been updated.

ERA5 wind stress at 25.5°N is now used to calculate the Ekman transport. This has resulted in an increase of the mean Ekman transport (0.8 Sv) compared with the previous data release when ERA-interim was used.

There were two mooring losses in the deployment from March 2017 to October 2018. One on the eastern boundary and one on the west. Because of the latter the time series has not been extended beyond 3 September 2018. Full details are given in the section “Data gaps”

One instrument on mooring EBH4 during the autumn 2015 to spring 2017 deployment has been identified as faulty and removed from the calculation of the eastern boundary temperature-salinity profile. The resulting change to the MOC during this period has a standard deviation of about 0.5 Sv.

A small improvement to the processing of data from mooring WB3 during the time from November 2005 to March 2006 has slightly increased (0.25 Sv) the MOC during that time. This was the time when WB3 was used to fill a gap in the record of WB2 (see “Data gaps” below). For some deployments, salinity data from the deepest instrument on WB3 has been found to suffer from fouling and a correction using the climatological T-S relation data has been applied.

October 2017

The time series is extended to 28th February 2017.

The coefficients used for extrapolation above the shallowest instruments have been updated.

Previously unused temperature and salinity measurements from MicroCATs at depths of about 400m on EBH4 since October 2012 have been included.

Previous time series have had a 10-day low-pass filter applied. This leads to spurious values in the first and last 5 days of the time series. In this release the same filter is applied, but the first and last 5 days of the time series have been set to absent data.

Each time the time series is updated the series mean and standard deviation change. These statistics affect interpolation and quality control of the data and so cause some minor changes to the calculated MOC.

In the mooring gridded-data files we have added a flag to indicate where climatological data has been used to fill gaps in the data. These gaps typically occur between deployments, or when gridded data are removed during quality control.

June 2016

The time series is extended to 11th October 2015.

The coefficients used for extrapolation above the shallowest instruments have been updated.

December 2014

The time series is extended to 22nd March 2014.

The calculation of the MOC has been modified as follows:

- The temperature and salinity gradients used for the interpolation of moored instrument data onto a high-resolution grid are now based on the Hydrobase3 climatology
- The coefficients used for extrapolation above the shallowest instruments have been

updated leading to differences with the previous release of <0.1 Sv standard deviation.

These changes are detailed in McCarthy et al., 2015. The mean change from the previous release is a reduction of 0.2 Sv.

In addition an updated version of the Florida Straits transport time series was used.

May 2013

The time series is extended to October 2012

The calculation of the MOC has been updated to:

- Change equation of state to TEOS-10
- Improve extrapolation of near surface data and use seasonal climatology
- Modify the transport of AABW
- Change the wind product from CCMP to ERA interim

Description of the data

Five data files are provided here:

1. moc_transports (available in Matlab, NetCDF and ASCII format)
2. moc_vertical (available in Matlab format)
3. ts_gridded (available in NetCdf format)
4. rapid_2D_data (NetCDF)
5. rapid_transport_timeseries (NetCDF)

All times are in days since 1st April 2004 unless stated otherwise.

The order of variables in the ascii file is:

```
JD  YY  MM  DD  HR
t_therm10  t_aiw10  t_ud10,  t_ld10  t_bw10
t_gs10     t_ek10   t_umo10  moc_mar_hc10
```

See below for a description of the variables.

Original instrument records from the moorings are available from the BODC.

1. MOC transport, components transports and transports in layers

The file *moc_transports* contains a 19.9-year time series of

- MOC transports (moc_mar_hc10)

for the period 2nd April 2004 to 27th March 2024 and the components:

- Upper mid-ocean transport (t_umo10),
- Florida Straits transport (t_gs10)
- Ekman transport (t_ek10).

Layer transports are also given for

- 0-800 m, “thermocline recirculation” (t_therm10)
- 800 to 1100m “intermediate water” (t_aiw10),
- 1100 to 3000m “upper North Atlantic Deep Water “ (t_ud10),
- 3000 to 5000m “lower North Atlantic Deep Water” (t_ld10),
- > 5000m “Antarctic Bottom Water” (t_bw10).

Negative transports correspond to southward flow. Transports are given in Sv ($10^6 \text{ m}^3\text{s}^{-1}$),

low-pass filtered at 10-day cutoff at twice daily resolution. The dates in the ASCII file are given in both Julian days and the Gregorian calendar. Julian day 1 corresponds to 2nd April 2004 at 00:00:00.

In order to calculate a maximum overturning, the methods of Cunningham et al. (2007) are followed, the upper mid-ocean transport is computed in a way that minimises the amount of southward transport, thus maximising the overall amount of northward transport of upper waters when Florida Straits, Ekman and mid-ocean transports are added together. The mid-ocean flow is generally southward from the surface to about 800 m depth, but between about 800 m and 1100 m depth there is a region of northward flow, while below about 1100 m depth the deep flow is generally southward. To obtain the minimum value of southward upper midocean transport at each time, the depth above which there is intermediate northward flow is found and below which there is southward flow. The upper mid-ocean transport is then defined to be the transport from the surface down to this depth at each time. On days where there is no northward flowing intermediate water and the mid-ocean transport is summed down to only 800 m because this is the approximate depth of the Florida Straits, where northward flowing Gulf Stream waters extend to the bottom.

Florida Straits transport

Florida Straits transport is based on electromagnetic cable measurements (Baringer & Larsen 2001). The Florida Current cable and section data are made freely available on the Atlantic Oceanographic and Meteorological Laboratory web page :

(<https://www.aoml.noaa.gov/phod/wbts/data.php>) and are funded by the NOAA Office of Climate Observations. Corrections for the secular change in the geomagnetic field were applied in 2024 (Volkov et al., 2024). A gap in the time series of approximately two months from 4th September to 28th October 2004 is due to hurricane Frances that destroyed the facility recording the voltage. Here linear interpolation is used to fill the gap.

Ekman transport

Ekman transport is calculated from the ERA5 wind stress (C3S, 2017) at 26.5°N. In earlier versions of the timeseries other data products were used, see the above change log for further details.

Negative values of the MOC

Negative MOC values: From 20th – 24th December 2009 and on 6th January 2010, there was no northward transport of water across 26.5°N. Usually, when there is no northwards transport, the MOC is seen to have two depth modes. If northward flow of Antarctic Intermediate Water is present in the geostrophic transport per unit depth, the MOC depth is 1050m. This is the case in all but one of the anomalous profiles. If no northward flow of AAIW exists, then the Florida Straits transport dominates and the MOC depth is 655m. This is the case in one of the anomalous profiles. Thus for the anomalous profiles when no northward flow exists, the MOC is defined as the cumulative vertical integral of all transports either from the surface to 1050m, if northward flow of AAIW is present in UMO, or to 655m if there is no northward flow of AAIW in UMO. Full details are given in the supplementary online material for McCarthy et al. 2012.

2. Streamfunction

The file *moc_vertical* contains the vertical profile of the overturning stream function in depth space at each time.

3. Gridded Mooring Data

The file `ts_gridded` contains the temperature and salinity data used in the calculation of the MOC. To calculate the interior geostrophic flow data from the different moorings are combined to create four merged profiles of temperature and salinity, one each for the:

- the western boundary (TG_west, SG_west)
- the western Mid Atlantic Ridge, (TG_marwest, SG_marwest)
- the eastern Mid Atlantic Ridge (TG_mareast, SG_mareast)
- the eastern boundary. (TG_east, SG_east)

Each of these merged profiles is then mapped onto a regular depth grid using the methodology of Johns et al. (2005). The resulting gridded profiles are supplied in the file `gridded_TS_profiles.mat`.

Data gaps

In the western boundary, the primary mooring WB2 was not in the water from 25th February 2023 to the 19th July 2023. For this period, the calculation of transport in the Western Boundary Wedge was extended out to WB3 and WB3 temperature and salinity data were used instead of WB2.

Data in the lower (3500 to 4500 m) of the Eastern boundary were lost between 1st September 2019 and 15rd March 2020. The mooring lost buoyancy, but it was still recovered.

Data in the upper c. 800m of the western boundary (mooring WB2) were lost from 3rd September 2018 to 21st November 2018. This followed damage to the mooring from fishing lines.

Data in the upper 1000m of the eastern boundary were lost between 1st March 2017 and 23rd October 2018. Mooring EBH4 was lost to fishing and so the backup mooring EBH3, 21 km west of EBH4 was used to construct the upper 1000m of the eastern boundary profile.

Comparison of data from previous deployments suggests that this may have a small impact on the seasonal cycle (± 0.5 Sv) but no significant impact on the mean value of the AMOC.

Data from the top 800 m of the mooring on the west flank of the MAR was lost from 10th October 2016 to the 28th February 2017 due to fishing activity. Data for this time period were replaced with a climatology based on previous deployments of the mooring.

From February to June 2006, data were lost on the eastern boundary due to a change in the firmware of the instruments. This gap was linearly interpolated over.

Data from the top 800m of the mooring on the west flank of the MAR was lost from 25th October to the 22nd December 2010 due to fishing activity. Data for this time period were replaced with a climatology based on previous deployments of the mooring.

The mooring on the eastern flank of the MAR was lost for the period 1st November 2009 to the 22nd December 2010. This time period was also replaced with a climatology based on previous deployments of this mooring.

In the western boundary, the primary mooring WB2 was lost from 7th November 2005 to the 26th March 2006. For this period, the calculation of transport in the Western Boundary Wedge was extended out to WB3 and WB3 temperature and salinity data were used instead of WB2.

4. Rapid 2D-data

The variables are:

- time - Time in days since 1950-01-01
- longitude, the longitudes of the centre of the grid cells.
- depth - depth at the centre of the grid cells
- pressure - the corresponding pressures

- area – the area of each grid cell
- SA – absolute salinity
- CT – conservative temperature
- vel_ekman – Northward Ekman velocity
- vel_insitu – the non-Ekman northward velocity
- vel_net – a uniform meridional velocity which is derived assuming that the salinity flux at 26.5°N is equal to a constant 33.4 Sv psu the time mean measured transport at the Bering Strait since 2004.

Note that:

- the Florida Strait is represented by a single column of data and the area varies with depth. Values are derived from the cable measurements and hydrographic sections
- the next three columns, vel_insitu(:,2:4,:), represent the 'western boundary wedge' where measurements are made with current meters. Area varies with depth.
- the remaining columns of velocities are calculated from geostrophy as described below

The methodology is very similar to that of Johns et al., (2023) and details are given in Smeed et al., (2026). East of the Western Boundary Wedge, variables are estimated on a grid with resolution 0.25 degrees of longitude from 76.5°W to 14°W and on a pressure grid of 20db resolution from the surface to 6120db. Above 1800db temperature and salinity are from an objective analysis of Argo and mooring data. Below 1800db and east of 74°W World Ocean Atlas (WOA) data (constant in time) are used. West of 74°W data are derived from the WBTS CTD sections. Optimal interpolation of WBTS data in time is made using a Gaussian weight function with a time scale of 1 year.

Dynamic height is calculated at each longitude with a reference level of 800db between 76.5°W and 74°W, this was chosen so that the transports in Antilles Current and Deep Western Boundary Current are consistent with previous estimates. East of 74°W the reference level is 1720db, or the bottom when the water depth is shallower.

Transport is determined by multiplying the sum of 'vel_insitu' and 'vel_ekman' by the area. A zonally uniform adjustment is added at each level east of 74°W so that the profile of transport matches that determined by the endpoint methodology, this also ensures that the total net transport from 'vel_insitu' and 'vel_ekman' is zero. Finally, a spatially uniform net transport is determined (McDonagh et al., 2015) by requiring that the salinity transport equals 33.4 Sv psu, the time mean measured transport at the Bering Strait since 2004.

5. Rapid transport timeseries

The MOC timeseries and stream function are produced in both depth and density coordinates. The MOC and stream function in depth coordinates in this file are the same as in 'moc_transports' and 'moc_vertical' files, but down sampled to once every 10-days. The meridional transports for heat and freshwater variables are derived from the data in 'rapid_2D_data'.

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